

PET UNIAXIAL GEOGRID

Polyester (PET) type geogrids are made from a woven matrix of high-tenacity polyester fibers coated with a protective polymer coating. The polyester fibers are oriented in one principal strength direction. Geogrids interlock with the soil providing long term reinforcement and are the most cost-effective method of creating additional horizontal space.

July 2024		Uniaxial Geogrid – Metric Values									
Material Properties		ASTM	2XT ³	3XT	5XT	7XT	8XT	10XT	20XT	22XT	24XT
	Wide Width (kN/m) MD ⁴	D6637 Ultimate	33.6	53.3	68.6	91.9	110.9	148.8	233.4	306.4	408.5
	Creep Rupture Strength(kN/m) ^{1,5}	D5262/D6992	23.2	35.5	47.6	55.3	77.0	103.3	167.2	212.8	283.7
	LTDS (kN/m) ^{2,5}	75-year design life – sand, silt, clay. RF _{CR} = 1.44; RF _{ID} = 1.05; RF _D = 1.1									
	Sand, Silt, Clay ^{2,5}		20.2	32.0	41.2	55.3	66.7	89.5	140.4	184.2	245.6
	Roll Width (m)		3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6
	Roll Length (m)		46	46	46	61	46	61	61	61	61
Roll Weight(kg)	49		52	52	52	93	102	157	226	270	

Note¹ 75-year design life based on NTPEP Report for each specific product – report available upon request.

Note² Value based on Long Term Design Strength for sand, silt, clay RF_{CR} = 1.44; RF_{ID} = 1.05; RF_D = 1.1

(Installation damage reduction factor for other soils available upon request)

Note³ Values shown for 2XT are both machine and cross-machine direction. Values for other products are machine direction only.

Note⁴ MARV – Minimum Average Roll Value

Note⁵ Minimum Roll Value

Uniaxial Geogrid – Imperial (US) Values											
July 2024											
Material Properties		ASTM	2XT ³	3XT	5XT	7XT	8XT	10XT	20XT	22XT	24XT
	Wide Width (lbs/ft) MD ⁴	D6637 Ultimate	2300	3650	4700	6300	7600	10200	16000	21000	28000
	Creep Rupture Strength (lbs/ft) ^{1,5}	D5262 D6992	1597	2535	3264	4375	5278	7083	11111	14583	19444
	LTDS (lbs/ft) ^{2,5}	75-year design life – sand, silt, clay. RF _{CR} = 1.44; RF _{ID} = 1.05; RF _D = 1.1									
	Sand, Silt, Clay ^{2,5}		1383	2195	2826	3788	4570	6133	9620	12626	16835
	Roll Width (ft)		12	12	12	12	12	12	12	12	12
	Roll Length (ft)		150	150	150	200	200	200	200	200	200
	Roll Weight Typical (lbs)		109	115	135	188	205	223	387	499	595

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